

A FOUR-VERTEX COUNTEREXAMPLE TO THE CLOW–VAN BOMMEL ETERNAL DISTANCE- k DOMINATION CONJECTURE

ABSTRACT. Let $\gamma_k(G)$ be the distance- k domination number of G and let $\gamma_{\text{all},k}^\infty(G)$ be its eternal distance- k domination number in the all-guards-move model. Conjecture 8.2 of Clow and van Bommel asserts that for all $k > 2$ and all trees T , if $\gamma_k(T) = \gamma_{\text{all},k}^\infty(T)$ then $\gamma_{\text{all},k}^\infty(T) = \gamma_{\lfloor k/2 \rfloor}(T)$. The path P_4 at $k = 3$ refutes it: there $\gamma_3 = \gamma_{\text{all},3}^\infty = 1$ while $\gamma_{\lfloor 3/2 \rfloor} = \gamma = 2$. The mechanism is parity, so the statement is false for every odd $k \geq 3$, witnessed by every tree of diameter exactly k . The value $\gamma_{\text{all},k}^\infty(T) = 1$ for $\text{diam}(T) \leq k$ is Lemma 3.3 of the same paper, so no new mathematics is claimed. Even k is not addressed.

1. THE CONJECTURE

All graphs are finite and simple, and $d(u, v)$ denotes graph distance. For $k \geq 1$ a set $S \subseteq V(G)$ is *distance- k dominating* if every vertex of G lies within distance k of a vertex of S ; the least size of such a set is $\gamma_k(G)$, and $\gamma_1(G) = \gamma(G)$ is the ordinary domination number. Write $N_k[v] = \{u : d(u, v) \leq k\}$.

Following Cox, Meger and Messinger [2], a family $\mathcal{F}$ of g -element multisets of $V(G)$ (*configurations*) is an *eternal distance- k dominating family* in the all-guards-move model if

- (i) every $A \in \mathcal{F}$ is distance- k dominating, and
- (ii) for every $A \in \mathcal{F}$ and every $v \in V(G)$ there are $B \in \mathcal{F}$ with $v \in B$ and a bijection of multisets $\sigma : A \rightarrow B$ with $d(a, \sigma(a)) \leq k$ for all $a \in A$.

The least such g is $\gamma_{\text{all},k}^\infty(G)$. Condition (i) — that *every* configuration of the family, not merely the initial one, is itself distance- k dominating — is the published definition [2, §2], and it is the reading used throughout. The refutation does not depend on this choice: for the families exhibited below, every configuration of the family is reachable, so the weaker reading — requiring only reachable configurations to distance- k dominate — gives the same values. Cox, Meger and Messinger also prove the sandwich

$$(1) \quad \gamma_k(G) \leq \gamma_{\text{all},k}^\infty(G) \leq \gamma_{\lfloor k/2 \rfloor}(G).$$

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Clow and van Bommel [1] prove for $k = 2$ that the two outer quantities in (1) coincide whenever the two left ones do, and in the Conclusion of that paper conjecture that the same holds beyond $k = 2$:

Conjecture 8.2 [1]. *For all $k > 2$ and trees T , if $\gamma_k(T) = \gamma_{\text{all},k}^\infty(T)$, then $\gamma_{\lfloor k/2 \rfloor}(T) = \gamma(T)$.*

The statement is quoted from the e-print [arXiv:2308.00054v1](#). That the quotation is faithful, and that no standing hypothesis of [1] has been dropped in transcribing it, is a documentary matter checked by reading the source, not by the program of §3.

2. THE COUNTEREXAMPLE

Theorem 1. *Let $k = 3$ and $T = P_4$, the path $1 - 2 - 3 - 4$. Then*

$$\gamma_3(P_4) = \gamma_{\text{all},3}^\infty(P_4) = 1, \quad \gamma_{\lfloor 3/2 \rfloor}(P_4) = \gamma(P_4) = 2.$$

Hence the hypothesis of Conjecture 8.2 holds and its conclusion fails, so Conjecture 8.2 is false.

Proof. $\text{diam}(P_4) = 3 = k$, so every vertex of P_4 is within distance 3 of every other; each of the four singletons is therefore a distance-3 dominating set and $\gamma_3(P_4) = 1$. The same computation makes the family $\mathcal{F} = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ of all four singletons eternal: each member distance-3 dominates, and an attack at v is answered by moving the lone guard to v , a move of length at most 3. So $\gamma_{\text{all},3}^\infty(P_4) = 1$, the value 0 being impossible. Finally $\lfloor 3/2 \rfloor = 1$, and no single vertex of P_4 dominates it: $N_1[1] = \{1, 2\}$, $N_1[2] = \{1, 2, 3\}$, $N_1[3] = \{2, 3, 4\}$, $N_1[4] = \{3, 4\}$. Since $\{2, 3\}$ dominates, $\gamma(P_4) = 2$. The conclusion of Conjecture 8.2 would require $1 = 2$. $\square$

Write $\text{rad}(T)$ for the radius; in a tree $\text{rad}(T) = \lceil \text{diam}(T)/2 \rceil$.

Proposition 2. *Let $k \geq 3$ be odd and let T be any tree with $\text{diam}(T) = k$. Then $\gamma_k(T) = \gamma_{\text{all},k}^\infty(T) = 1$ while $\gamma_{\lfloor k/2 \rfloor}(T) \geq 2$, so T is a counterexample to Conjecture 8.2 at that k .*

Proof. If $\text{diam}(T) \leq k$ then a single vertex distance- k dominates T and every move is legal, so exactly as in Theorem 1 $\gamma_k(T) = \gamma_{\text{all},k}^\infty(T) = 1$; the eternal value is also the second case of Lemma 3.3 of [1], proved there five sections before Conjecture 8.2. Next $\gamma_j(T) = 1$ if and only if $\text{rad}(T) \leq j$. With $\text{diam}(T) = k$ odd, $\text{rad}(T) = (k+1)/2 > (k-1)/2 = \lfloor k/2 \rfloor$, so $\gamma_{\lfloor k/2 \rfloor}(T) \geq 2$ and the conclusion fails. $\square$

The obstruction is parity: $2\lfloor k/2 \rfloor + 1$ equals $k+1$ for even k but only k for odd k . The least tree of diameter exactly k is P_{k+1} , which gives Theorem 1 at $k = 3$.

Scope. Conjecture 8.2 is universally quantified over $k > 2$, so a single odd k refutes it as written. The mechanism needs k odd, and nothing above bears on the restriction of Conjecture 8.2 to even k , or on the other conjectures and

questions of [1, §8]. We refute the printed statement and do not speculate about intent.

Attribution. Theorem 1 is Lemma 3.3 of [1] — quoted, not reproved — together with $\gamma(P_4) = 2$; the observation that these contradict Conjecture 8.2, with the parity account of it, is what this note adds.

3. VERIFICATION

Everything above is checkable by hand from the objects printed here: P_4 , and P_{k+1} for Proposition 2. The accompanying program `verify.py`, with its recorded transcript `verify.output.txt`, re-derives the displayed numbers independently — distances, radii and diameters by breadth-first search, each γ_j by an exact minimum set cover, and each $\gamma_{\text{all},k}^\infty$ by a greatest-fixed-point solution of the game of §1 over its full configuration space. It uses exact integer arithmetic and the Python standard library only.

The transcript is that of an earlier and wider draft, and the reader should read it accordingly. Its Steps 1, 2 and 4 are the checks that belong to this note: Step 1 and Step 2 verify every number in Theorem 1, and Step 4 instantiates Proposition 2 at $k = 3, 5, 7$ and on one non-path tree, the general statement resting on the proof above and not on those instances. Everything else in the transcript concerns material this note does not contain and does not claim: Step 3 treats P_8 at $k = 3$; Steps 5 and 6 are headed “Corollary 4” and “Corollary 5” and refer to a path family, to published closed forms and to a correction to [2], none of which is stated here; Step 7 is a bounded sweep of the 95 trees of order at most 9; and the closing scope notes mention a wider census of trees, and locations inside the source tarball, that this note does not cite. The internal numbering of the transcript is that of the earlier draft and does not match the numbering printed above. The program checks no statement about the contents of [1] or [2]; those are documentary claims, checked by reading.

REFERENCES

- [1] A. Clow and C. M. van Bommel, *Eternal distance-2 domination in trees*, *Ars Combinatoria* **163** (2025), 29–49. doi:10.61091/ars163-03; arXiv:2308.00054v1. Conjecture 8.2 and Lemma 3.3 are quoted from the e-print.
- [2] D. Cox, E. Meger, and M. E. Messinger, *Eternal distance- k domination on graphs*, *La Matematica* **2** (2023), no. 2, 283–302. doi:10.1007/s44007-023-00044-3; arXiv:2104.03835. Source of the definition in §1 and of the sandwich (1).